CSE Undergraduate Project Ethics Declaration

Research ethics helps us identify and address potential risks of harm to participants, researchers and others affected by a project. This ethics declaration adapts the self-assessment questions in the University's standard ethics application form for CSE undergraduate final projects. Supervisor acceptance is recorded separately. For the University's ethics policy and guidance, see Research and Knowledge Exchange Ethics.

Undergraduate projects are normally expected to fall within Class 1, i.e. projects with low or no risk of harm. Projects requiring Class 2 or higher review must follow the formal University ethics application and approval process. Exceptionally, your supervisor may agree to support such a project and submit the ethics application on your behalf.

This form is not intended for projects involving animal research, animal tissue, human tissue or the collection of human DNA samples. If your project involves any of these, discuss the appropriate University ethics and governance process with your supervisor.

1. Student and project details

Student full name

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Module

6SENG010W Software Engineering Final Project

Supervisor full name

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Project title

Market-State-Conditioned Orchestration of Heterogeneous Agents for Financial Decision Support: A Colombo Stock Exchange Study

Project start date

01/08/2026

Project end date

01/04/2027

Brief project description and methodology

Summarise the aims, activities, evaluation and data (maximum 200 words).

Aim: To design, implement and evaluate a market-state-conditioned orchestration operator that continuously varies the influence of methodologically heterogeneous financial agents over a single shared decision, and to establish, on Colombo Stock Exchange data under a cost-aware and regime-decomposed protocol, under what conditions it earns its place against fixed, static learned and performance-reactive weighting.

Activities: Design science research, with the artefact built by evolutionary prototyping under stage gates. A provisional set of four agents, technical, fundamental retrieval, volatility and liquidity, runs behind one output contract and feeds an orchestrator that sets each agent's influence from measured liquidity, volatility, persistence and agent confidence. Pre-trained models are used through third-party APIs under their terms. Sentiment analysis is out of scope.

Evaluation: Offline retrospective simulation only, with no live trading and no deployment. Five weighting arms run over identical cached agent outputs on a rolling walk-forward split of sixteen folds, with per-input ablations, cost sensitivity sweeps, regime decomposition and significance testing. A null result is pre-registered as reportable.

Data: Historical Colombo Stock Exchange price, volume, corporate action and float records, 2012 to 2026, purchased under a non-commercial student licence, plus 22,893 public company filings. No human participants, no personal data, nothing redistributed.

Existing ethics approval

1.1 Is your proposed project fully covered by an existing, valid University of Westminster ethics approval? ○ Yes ○ No

Select Yes only if your supervisor has confirmed that the approval covers your proposed activities, participants, data, locations and timescale. External approval alone, or a Westminster application awaiting a decision, does not qualify.

If Yes: Ethics Application Reference

If Yes, proceed to section 8, Declaration outcome and confirmation. If No, complete the remaining sections as directed.

2. External factors

2.1 Does your research include funding from an external organisation and/or involve external collaborators or co-investigators? ○ Yes ○ No

'External organisation' refers to any bodies which: conduct, host, sponsor or fund research; employ, support or host researchers; teach research students; or allow research to be carried out under their auspices.

Include payment from an organisation to undertake your project, funding applications not yet awarded, and collaboration with or without a financial agreement. Ordinary student finance and employment unrelated to the project do not count.

For Industry Exchange Programme (IXP) projects, answer No if the only external involvement is the standard industry collaboration already screened and established through CSE. Answer Yes if your project involves additional external funding, collaborators or arrangements outside that agreed setup.

2.2 Does your project require, or have you sought or obtained, ethics approval or other formal research/governance permission from an external organisation? ○ Yes ○ No

Examples include the Health Research Authority, another university, prison authorities or the Ministry of Defence.

2.3 Have you been asked by an external organisation to produce evidence of ethical approval for your research? ○ Yes ○ No

Further details

If you answered Yes to any question above, please provide further details, identifying the organisations involved and the relevant arrangements or requirements.

3. Project scope

Answer all questions below about your proposed project.

3.1 Will people participate in your research or evaluation? ○ Yes ○ No

This includes interviews, surveys, observation, experiments and testing your software or other artefact. Friends and classmates taking part count as participants.

If Yes, complete sections **6 and 7**.

3.2 Will your project collect, access, analyse or otherwise use data; develop, train, fine-tune or evaluate an AI model; or incorporate an AI model into the project's analysis or proposed solution? ○ Yes ○ No

Include personal data, including data used to manage research participants. Information used solely for routine academic administration, such as supervision records, does not, by itself, require a Yes answer. Include data science, analytics, machine learning and AI projects. Include pre-trained models, whether run locally or accessed through third-party applications or APIs. Routine AI assistance, such as help with writing or coding, does not, by itself, require a Yes answer.

If Yes, complete section **5A: Data, analytics and AI**.

3.3 Does your project involve developing, using or evaluating games, interactive simulations or immersive technologies, or collecting or analysing data generated through their use? ○ Yes ○ No

Immersive technologies include VR, AR, MR and head-mounted displays.

If Yes, complete section **5B: Games, simulations and immersive technologies**.

3.4 Does the project involve security testing, digital forensics, malicious code or security-related investigation? ○ Yes ○ No

Ordinary use of security features, such as login or encryption, does not by itself require this section.

If Yes, complete section **5C: Cybersecurity and digital forensics**.

4. Fieldwork

4.1 Will any project activities be carried out in locations you would not normally use for University study or work? ○ Yes ○ No

Examples include visits to external organisations, fieldwork sites or locations used for participant activities. Routine work from home does not, by itself, count.

If Yes, identify the locations, the activities planned there and any access or permission arrangements.

Fieldwork locations, activities and permissions

5. Risk of harm

Complete the general questions below and the additional subsections indicated by your answers in section 3. If an activity described in a question is not part of your project, answer No.

5.1 Will any pain or more than mild discomfort result from the study? ○ Yes ○ No

5.2 Could the study induce any psychological stress or anxiety or cause harm or negative consequences beyond the risks encountered in normal life? ○ Yes ○ No

5.3 Will the study involve prolonged or repetitive physical or psychological testing of human participants that may put someone at risk, e.g. use of a treadmill? ○ Yes ○ No

5.4 Will the study involve raising sensitive topics, e.g. sexual activity, drug use, revelation of medical history, bereavement or illegal activities? ○ Yes ○ No

5.5 Does your study raise any issues of personal safety for you or other researchers or participants involved in the project? ○ Yes ○ No

Especially relevant if taking place outside working hours or off University premises.

5.6 Does your study involve deliberately misleading the participants, e.g. deception or covert observation? ○ Yes ○ No

5.7 Does your research involve any security-sensitive material? ○ Yes ○ No

See Universities UK, Oversight of Security-Sensitive Research Material (2019). Cybersecurity and digital-forensics activities covered by section 5C do not, by themselves, require a Yes answer here.

5A. Data, analytics and AI

Complete this subsection if you answered Yes to question **3.2**.

5A.1 Does your project collect, access, use or share identifiable or potentially re-identifiable personal data without appropriate safeguards? ○ Yes ○ No

Consider consent or other authorisation, confidentiality, secure storage, access restrictions, retention and disposal. Include identifiable source data handled before anonymisation. Routine participant contact details or low-risk feedback, handled with appropriate safeguards, do not by themselves require a Yes answer.

5A.2 Does your project collect, access or use identifiable or potentially re-identifiable personal data of a particularly sensitive nature? ○ Yes ○ No

Examples include health information, political or religious beliefs, sexual information, and criminal allegations or convictions. Include other confidential personal information whose disclosure could cause particular harm, distress or discrimination. Recordings count where they contain or reveal such information.

Answer Yes even where appropriate safeguards are proposed. Exclude genuinely anonymous information, but include sensitive identifiable source data handled before anonymisation.

5A.3 Does your project collect or use data, datasets, models or services in breach of applicable access or usage conditions, or without required permission or authorisation? ○ Yes ○ No

Consider dataset licences, scraping/API conditions, organisational authorisation and model/service terms. An explicit permission letter is unnecessary where the applicable terms already permit your proposed use.

5A.4 Does the project's proposed testing, deployment or live use of analyses or model outputs make or materially inform decisions about real people that affect their health, safety, access to services, opportunities or other significant interests? ○ Yes ○ No

Include live trials where outputs inform real decisions. Evaluation using historical or synthetic data does not, by itself, require a Yes answer.

5A.5 Could people reasonably rely on, or be affected by, the project's findings or model outputs during its proposed testing, use, publication or release in a way that causes harm or material disadvantage? ○ Yes ○ No

Consider incorrect or misleading outputs, bias or under-representation, unfair treatment, and misleading or stigmatising conclusions about groups. A dataset having limitations does not, by itself, require a Yes answer; consider their potential consequences in the proposed project.

5B. Games, simulations and immersive technologies

Complete this subsection if you answered Yes to question **3.3**.

5B.1 Does the study involve participant testing using VR, AR, MR, head-mounted displays or other immersive technologies in a manner that may present risks beyond normal or short-duration use, such as prolonged exposure, repeated sessions, significant physical movement, or an increased likelihood of cybersickness or physical discomfort? ○ Yes ○ No

5B.2 Does participant testing involve specialist gaming or immersive hardware in circumstances that introduce physical hazards beyond normal supervised use, such as an increased risk of collision, falls, restricted movement or injury? ○ Yes ○ No

5B.3 Does the study expose participants to intense visual, auditory or haptic stimuli, e.g. flashing/flickering images, loud or sudden sounds, jump scares, strong vibration or rapid motion, that could trigger adverse sensory or neurological reactions? ○ Yes ○ No

5B.4 Does the study intentionally expose participants to, or require them to engage with, game content or scenarios specifically designed to provoke significant fear, distress, anxiety or other strong emotional responses beyond those normally expected from age-appropriate commercial gameplay? ○ Yes ○ No

5B.5 Does the study collect or record gameplay, behavioural or biometric data from participants without their prior knowledge and informed consent (e.g. movement/body tracking, eye-gaze, facial expression, voice/audio, physiological signals, controller input, location or detailed interaction logs)? ○ Yes ○ No

The collection of such data for research purposes is normally acceptable where participants are clearly informed in advance about what will be recorded, how the data will be used and stored, and provide appropriate consent before participation.

5C. Cybersecurity and digital forensics

Complete this subsection if you answered Yes to question **3.4**.

5C.1 Does the project involve intrusive security testing on live systems, networks, applications or devices involving real data, users or operational processes? ○ Yes ○ No

Exclude testing confined to a dedicated environment containing only test accounts and synthetic data, with no operational use or effect on systems outside that environment.

5C.2 Does the project access, collect, analyse or store non-public digital material whose exposure or misuse could compromise a real person's privacy or safety, or the security of an organisation or operational system? ○ Yes ○ No

Examples include usable credentials, private communications, forensic images containing confidential information, or network captures containing sensitive data. Exclude synthetic data, dummy credentials and traffic generated solely within a controlled test environment, provided these contain no sensitive information about real people or operational systems.

5C.3 Does the project create, modify or execute functioning malware or malicious code capable of compromising systems, stealing data, damaging information or disrupting services? ○ Yes ○ No

Include work with functioning malware even where an isolated environment is proposed. Exclude static analysis without execution and harmless simulations that do not carry out those functions. Use of a standard security tool does not, by itself, require a Yes answer.

5C.4 Does the project produce or reveal information, code or other outputs that would enable unauthorised access to, compromise of, or harm real people, organisations or operational systems? ○ Yes ○ No

Exclude findings confined to deliberately vulnerable test systems where they do not expose or enable attacks on operational systems.

6. Participants

Complete this section if you answered Yes to question **3.1**.

Who will participate, how will they be recruited, and what will they do?

Include the expected number of participants, relevant characteristics such as age, how you will select and approach them, and the activities and time commitment involved (maximum 200 words).

Does your work involve any of the following?

6.1 Human participants in Health and Social Care settings? ○ Yes ○ No

6.2 Human participants who may be vulnerable due to their setting(s)? ○ Yes ○ No

6.3 Expectant or new mothers? ○ Yes ○ No

6.4 Refugees or asylum seekers or recent migrants? ○ Yes ○ No

6.5 Minors (under the age of 18 years old)? ○ Yes ○ No

6.6 Participants in custody (e.g. prisoners or arrestees)? ○ Yes ○ No

6.7 Participants who may potentially fall under the remit of the Mental Capacity Act? ○ Yes ○ No

Mental Capacity Act guidance (NHS)

7. Information to participants

Complete this section if you answered Yes to question **3.1**.

Provide the proposed participant information, consent material and tasks or questions with your declaration.

7.1 Will you provide participants with a Participant Information Sheet prior to obtaining informed consent? ○ Yes ○ No

7.2 Will you describe the procedures to participants in advance, so that they are informed about what to expect? ○ Yes ○ No

7.3 Will you obtain informed consent for participation (normally written)? ○ Yes ○ No

7.4 Will you tell participants that they may withdraw from the research at any time and for any reason? ○ Yes ○ No

7.5 Will you give participants the option of omitting questions they do not want to answer? ○ Yes ○ No

7.6 Will you tell participants that their data will be treated as confidential and that, if published, it will not be identifiable as theirs? ○ Yes ○ No

7.7 Will you offer feedback to participants at the end of their participation, upon request (e.g. give them a brief explanation of the study and its outcomes)? ○ Yes ○ No

8. Declaration outcome and confirmation

Complete all required questions before selecting an outcome. The declaration outcome is determined by your answers:

• If you answered **Yes to question 1.1**: **Existing ethics approval**

• Otherwise, if you answered **No to all applicable questions in sections 2, 5, 5A–5C and 6**, and **Yes to all questions in section 7 where applicable**: **Class 1 sign-off**

• Otherwise: **Ethics review required**

All outcomes are subject to supervisor acceptance, including their assessment of the project's risks of harm beyond the questions in this form.

Outcome

Class 1 sign-off

If this has not been filled automatically, select the outcome using the rules above.

Student confirmation (electronic signature)

By typing my name below, I confirm that my project falls within the scope stated in the introduction and that this declaration is complete and accurate to the best of my knowledge. I confirm that I have not undertaken research activities involving participants or other potential risks of harm without the required sign-off or approval, and will not do so until my supervisor has confirmed that these are in place. I will discuss any project changes that affect this declaration with my supervisor and submit a revised declaration promptly.

Full name

Mohammed Shaahid Fathul

Date

02/11/2026

Save the completed declaration and submit it via Blackboard.